LCMS Retention Modeling

Python-based analysis of LCMS retention time using RDKit-derived molecular descriptors

Project Overview

This project evaluated whether LCMS retention time could be predicted from RDKit-derived molecular descriptors and whether the resulting models would recover descriptor relationships consistent with established chromatographic theory. Because reversed-phase LC retention is known to be strongly influenced by lipophilicity and related polarity effects, the analysis focused on both predictive performance and scientific interpretability.

The dataset contains one row per molecule, with rt as the target variable and 123 molecular descriptor columns used as predictors.

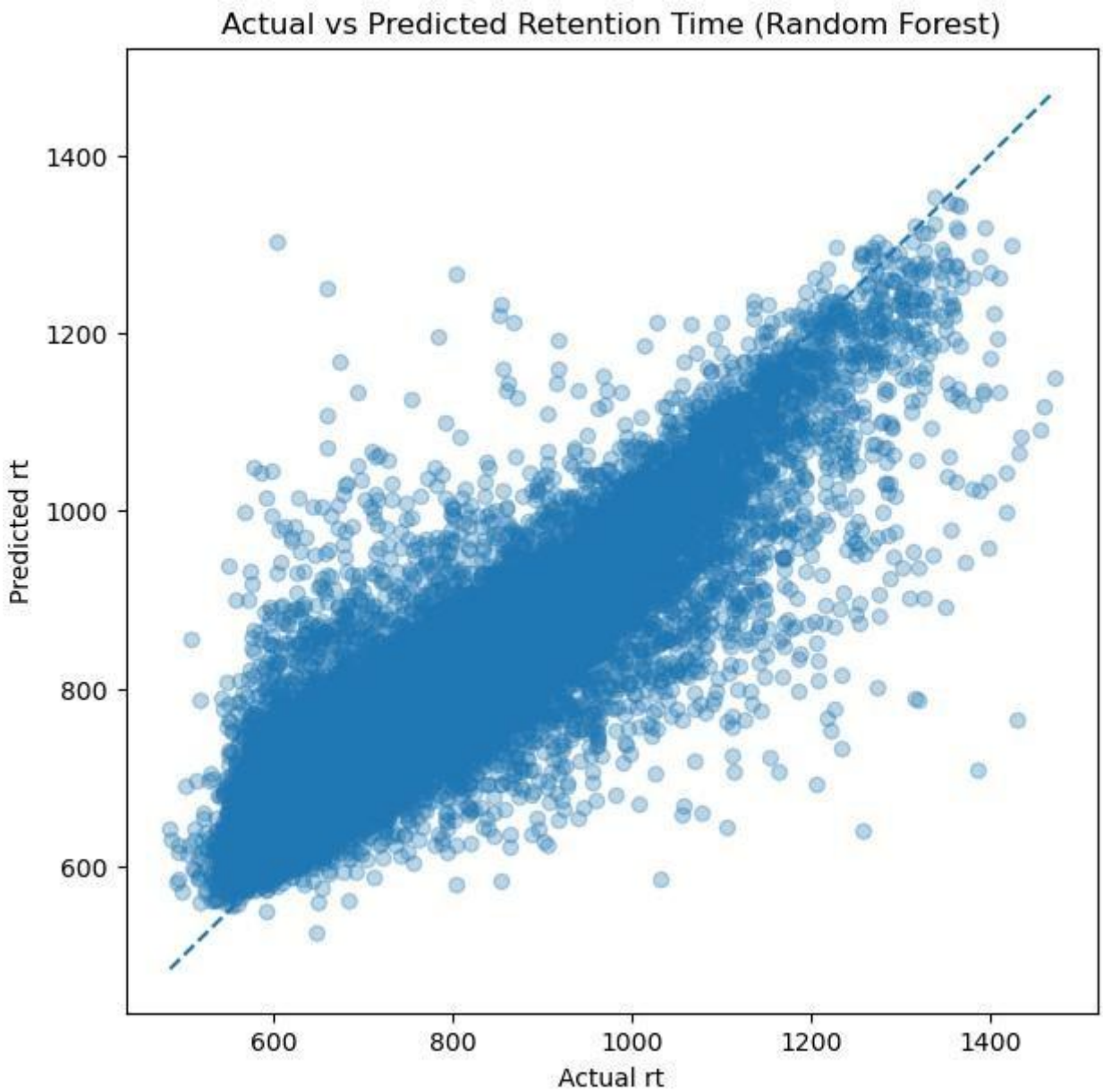
Workflow

- Loaded and inspected the dataset
- Performed data quality checks
- Explored the retention-time distribution
- Screened descriptor relationships using correlation analysis
- Established a mean baseline for comparison
- Fit Linear Regression and Ridge Regression baselines
- Trained an untuned Random Forest
- Tested Ridge and Random Forest tuning through cross-validated search
- Evaluated all models using MAE, RMSE, and R^2
- Performed error analysis across the retention-time range
- Interpreted coefficients and feature importances

Model Comparison

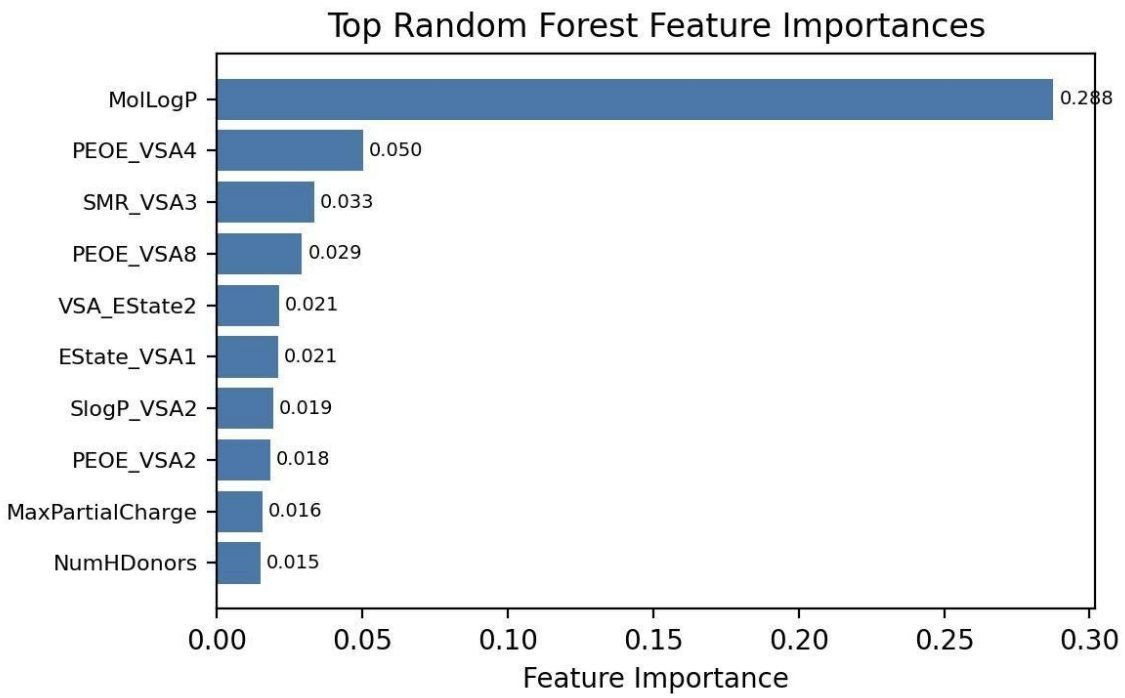
Model	MAE	RMSE	R^2
Mean Baseline	142.596	175.250	0.0000
Linear Regression	84.261	111.900	0.5923
Ridge Regression	85.306	112.926	0.5848
Ridge Regression (Tuned)	84.477	112.136	0.5906
Random Forest (Untuned)	58.665	85.441	0.7623
Random Forest (Tuned)	74.962	100.508	0.6711

Untuned Random Forest performed best, showing that non-linear descriptor relationships improved LC retention prediction and that the reduced tuning search did not improve held out performance.



Key Findings

- Retention time showed moderate predictability from the available 2D molecular descriptors.
- Linear and Ridge regression captured a meaningful but limited linear signal.
- The untuned Random Forest substantially improved predictive accuracy over all other tested models.
- Prediction error was lowest in the mid retention range and increased at both tails.
- The model systematically overpredicted low-retention compounds and underpredicted high-retention compounds.
- The strongest predictors were related to lipophilicity, surface area, charge distribution, and molecular size



Interpretation

Linear models were retained as interpretable baselines, even though Random Forest performed better overall. This allowed the project to compare broad linear descriptor effects with stronger non-linear predictive performance.

At a high level, the results support expected reversed-phase LC behavior: compounds with stronger lipophilic or nonpolar character and relevant surface-area patterns tend to show longer retention. MolLogP remained the strongest Random Forest feature, supporting the conclusion that lipophilicity-related descriptors dominate retention behavior in this dataset..

Limitations

- The dataset contains only 2D molecular descriptors and does not capture full structural or experimental complexity.
- Many descriptors are highly correlated, limiting one-to-one interpretation of model coefficients.
- The models are predictive and associative; they do not prove chromatographic mechanism.
- No experimental condition metadata was included beyond the descriptor matrix.
- Error increased at both tails of the retention-time distribution, especially for high-retention compounds.

Conclusion

The untuned Random Forest was selected as the final model because it achieved the strongest held-out test performance while also confirming the expected importance of lipophilicity-related descriptors. Error analysis showed that the model performs best in the mid retention range but is less stable at the extremes, especially for high-retention compounds. Overall, the project demonstrates that descriptor-based modeling can recover a substantial portion of LC retention behavior while also revealing where predictive performance remains limited.